Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Chanel
Project / Solution Name	KompasAI
Name (Leader)	Brayden Chong Jie Rui
Email & Phone (Leader)	braydencjr05@gmail.com 0125489958
University Affiliation(s)	Universiti Malaya
Live Prototype URL	https://kompas-ai.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/braydencjr/KompasAI.git

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Kompas has two parts joined by one Python FastAPI backend. The first part runs on its own: a signal pipeline collects records from public sources, including arXiv, GitHub, Hacker News, Malaysian news sites and Reddit, the Department of Statistics Malaysia, TalentCorp's Critical Occupations List, the European Skills, Competences, Qualifications and Occupations catalogue and the International Labour Organization's generative-AI exposure index. It tags each record to skills with a dictionary, counts them per month and places each skill on the Signal Ladder using fixed rules. Records, scores and dated snapshots are stored in Supabase Postgres, or a local SQLite file when offline. The second part is the user journey. In the Next.js web app, a user uploads a CV. The backend sends its text to Google Gemini 2.5 Flash, which lists the skills with quotes that code checks against the CV. Job scores come from rules over the stored data; Gemini writes the reasons and fills gaps where data is missing, shown as estimates. Anthropic Claude builds the work simulations. JSearch, Coursera, ImpactMojo and Tavily supply live jobs and courses. Accounts sit in a separate SQLite file. Replit hosting is planned; the prototype runs locally today. See Appendix A.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.11+	Backend, signal pipeline and tests	Python Software Foundation License
TypeScript 5.9.3	Frontend language	Apache-2.0
Next.js 15.5.28	Frontend web framework	MIT
React 19.1.0 and React DOM 19.1.0	User interface components	MIT

Tailwind CSS 4.3.3 (with @tailwindcss/postcss)	Styling	MIT
PostCSS 8.5.31	CSS build step	MIT
Geist, Geist Mono and Newsreader fonts (Google Fonts, via next/font)	Typography	SIL Open Font License 1.1
Google Identity Services JavaScript library	"Sign in with Google" button	Google APIs Terms of Service
FastAPI 0.154.0 (with Starlette and Pydantic 2)	Backend REST API and live feed (server-sent events)	MIT (Starlette: BSD-3-Clause)
Uvicorn 0.54.0	Web server for the backend	BSD-3-Clause
HTTPX 0.28.1	HTTP client for every data connector	BSD-3-Clause
Requests 2.34.2	HTTP client in the backend app	Apache-2.0
pypdf 6.21.1	Reading text from PDF CVs	BSD-3-Clause
openpyxl 3.1.5	Reading Department of Statistics Malaysia spreadsheets	MIT
psycopg 3 and psycopg-pool (3.2 or later)	Connecting to Supabase Postgres	LGPL-3.0 (used unmodified)
SQLite (bundled with Python)	Local database for offline use, accounts and AI answer cache	Public domain
python-multipart 0.0.32	File uploads	Apache-2.0
python-dotenv 1.2.4	Loading settings from the .env file	BSD-3-Clause
google-auth 2.62.0	Verifying Google sign-in tokens	Apache-2.0
Anthropic Python SDK 2.1.0	Calling the Claude API	Apache-2.0
Google Gen AI SDK (google-genai) 2.32.0	Calling the Gemini API	Apache-2.0
pytest 9.1.1	Automated tests	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Gemini API (Google)	Reads CVs, reads text from CV images, writes job reasons and learning plans, estimates values where data is missing, tags SDGs, rewrites AI Live Trend headlines, and creates skill embeddings	Paid
Claude API (Anthropic)	Generates work simulations and grades the "Direct the AI" task	Paid, pay as you go
JSearch API (OpenWeb Ninja, via RapidAPI)	Live Malaysian job openings and the skills employers ask for	Free plan
Tavily Search API (Tavily)	Finds free courses on approved learning sites	Free plan
Coursera Catalogue API (Coursera)	Course list for learning routes	Free, no key
ImpactMojo Courses API (ImpactMojo)	Free development-education courses for learning routes	Free, no key
arXiv API (Cornell University)	AI research paper counts (Signal Ladder rung 1)	Free
GitHub REST API and GitHub Trending page (GitHub)	Open-source activity and trending repositories	Free
Hacker News Search API (Algolia) and Hacker News API (Y Combinator)	Tech news and "Who is hiring?" posts (rungs 1 and 2), live feed	Free
Reddit API and public RSS feeds (Reddit)	Posts from r/malaysia and r/MalaysianPF for AI Live Trend	Free

RSS feeds of Malay Mail, Free Malaysia Today, Vulcan Post, SoyaCincau, Lowyat.NET and Digital News Asia	Malaysian news headlines and links for AI Live Trend	Free
ESCO API and dataset download (European Commission)	Skills and occupations catalogue	Free
Department of Statistics Malaysia website	Job Market Insights releases (rung 3)	Free
Google Identity Services (Google)	Sign in with Google	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Supabase Postgres (Supabase)	Main database: raw records, tagged evidence, Signal Ladder scores, dated snapshots, ESCO catalogue, cached AI answers	Free plan
Google AI Studio (Google)	Hosts the Gemini API project and key	Free tier
Anthropic Console (Anthropic)	Hosts the Claude API account and key	Pay as you go
Google Cloud Console (Google)	Planned OAuth client ID for Sign in with Google.	Free
GitHub (GitHub)	Source code repository	Free
Replit (Replit)	Hosted for the frontend and backend.	Planned

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Gemini 2.5 Flash (gemini-2.5-flash)	Google, via the Gemini API	Pre-trained, used with prompts and JSON schemas; no fine-tuning	Gemini API Additional Terms of Service; Generative AI Prohibited Use Policy
Gemini Embedding (gemini-embedding-001, 256 dimensions)	Google, via the Gemini API	Pre-trained; matches skills by meaning	Gemini API Additional Terms of Service
Claude Opus 5.5 (claude-opus-5-5, set in ANTHROPIC_MODEL), with Anthropic's server-side fallback model if a request is declined	Anthropic, via the Claude API	Pre-trained, used with prompts and structured outputs; no fine-tuning	Anthropic Commercial Terms of Service; Usage Policy

4.2 Datasets

Dataset Name	Source / URL	Type	License	Personal Data? (Y/N)
International Labour Organization Generative AI exposure	ilo.org/node/706766; pgmyrek.github.io/2025_GenAI_scores_ISCO08	Public	CC BY 4.0	N

index (2025), task-level scores				
ESCO classification v1.2.1 (13,939 skills and occupations)	esco.ec.europa.eu	Public	CC BY 4.0	N
ISCO-08 / MASCO 2020 occupation codes	International Labour Organization; Department of Statistics Malaysia	Public	Free reuse with attribution	N
Department of Statistics Malaysia Job Market Insights (quarterly)	dosm.gov.my	Public	DOSM open data terms, with attribution	N
MyCOL Critical Occupations List editions (TalentCorp and the Critical Skills Monitoring Committee)	talentcorp.com.my (PDF reports, imported by hand)	Public	Public government report, with attribution	N

arXiv paper metadata	export.arxiv.org/api	Public	CC0 1.0 (metadata)	Y (public author names only)
GitHub repository metadata and Trending page	api.github.com; github.com/trending	Public	GitHub Terms of Service	N (owner details not kept)
Hacker News stories and "Who is hiring?" posts	hn.algolia.com/api; hacker-news.firebaseio.com	Public	Hacker News API terms	N
Malaysian news headlines (RSS) and Reddit posts	Publisher RSS feeds; reddit.com	Public	Publisher terms; Reddit Data API Terms (headline and link only)	N
Job postings from JSearch	openwebninja.com / RapidAPI	Public	JSearch Terms of Service	N
Course listings from Coursera, ImpactMojo and Tavily search	api.coursera.org; impactmojo.in; api.tavily.com	Public	Provider terms; ImpactMojo CC BY-NC-ND 4.0 (link only)	N

Kompas curated tables (skills, aliases, occupation tasks, SDG tags, prerequisites, resources)	This repository (backend/pipeline/curated)	Self-collected (some rows drafted by Gemini from ESCO and reviewed by the team)	Apache-2.0 (team's own)	N
Sample CV (fictional person "Aisyah Rahman")	This repository (frontend/public)	Synthetic	Apache-2.0 (team's own)	N
CVs uploaded by users at run time	Users	Self-collected	N/A	Y
User accounts and saved progress	Users	Self-collected	N/A	Y

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

✅ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code (Anthropic), running Claude Opus models	Writing and refactoring code, tests, documentation, submission text, pitch deck and diagrams	Major

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

A team member read every change Claude Code proposed before committing it, and we ran and clicked through each feature in the app before merging. The backend has an automated pytest suite covering every data connector, the scoring rules, the CV reader and sign-in, run before each merge. Where the app itself calls AI models, code checks the output: CV quotes must appear word for word, estimates must cite real records, and simulation answers are verified with formulas and reference solutions. The team takes full responsibility for all submitted code and text.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

✅ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit		• Deployment hosting • UI improvement suggestion

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

It is a core functionality to the prototype as we use it as hosting , at the same time , technical wise it can be replaced by other hosting services.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

A CV is collected only when uploaded. Its text is read in memory, sent to the Google Gemini API to list skills and past jobs, and never stored on our server; the model is told not to return the person's name or contact details. The resulting skills profile stays in the user's browser. For a work simulation, only the role title, skill names and skill gaps go to Anthropic's Claude API, with no name or CV text. Users who create an account give an email, a display name and a password, or sign in with Google. These and saved progress sit in a backend SQLite file. Passwords are stored only as salted scrypt hashes, and sign-in tokens are stored as hashes and expire after 30 days. Accounts are kept until we delete them on request; a self-service delete option is planned. API keys stay in environment variables, never in code.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Gemini can misread a CV or invent skills, so every skill and job it returns must quote the CV word for word and match Kompas's own skill and job lists, or it is dropped. Where data is silent, Gemini estimates task exposure and skill demand; these estimates are checked in code against the records they cite, never replace measured data, and are labelled as estimates. Job scores come from fixed rules, so the same data always gives the same score. Sources lean towards English-language, technology and online job ads, so informal and non-technical work is under-counted; each source states its coverage limits. Free-tier rate limits on Gemini, JSearch and Tavily are eased by caching answers. If a model or source fails, Kompas falls back to the skills dictionary, the last good snapshot or built-in sample simulations. Simulations can run on Claude or Gemini, reducing dependence on one provider.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ✅ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ✅ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ✅ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ✅ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ✅ Any use of official SDC sponsor resources has been declared in Section 5.
- ✅ The prototype described here is accessible to judges through the links provided in Section 1.
- ✅ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



Date: 25/10/2026